



KIET
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Connecting Life with Learning



SYNOPSIS Presentation
on
Cybersecurity Visual Demo

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Cyber Security Visual Demo

An interactive educational project for cybersecurity fundamentals.



Demystifying Network Security

Visual Simulation

Understand networking and cryptography through interactive demonstrations.

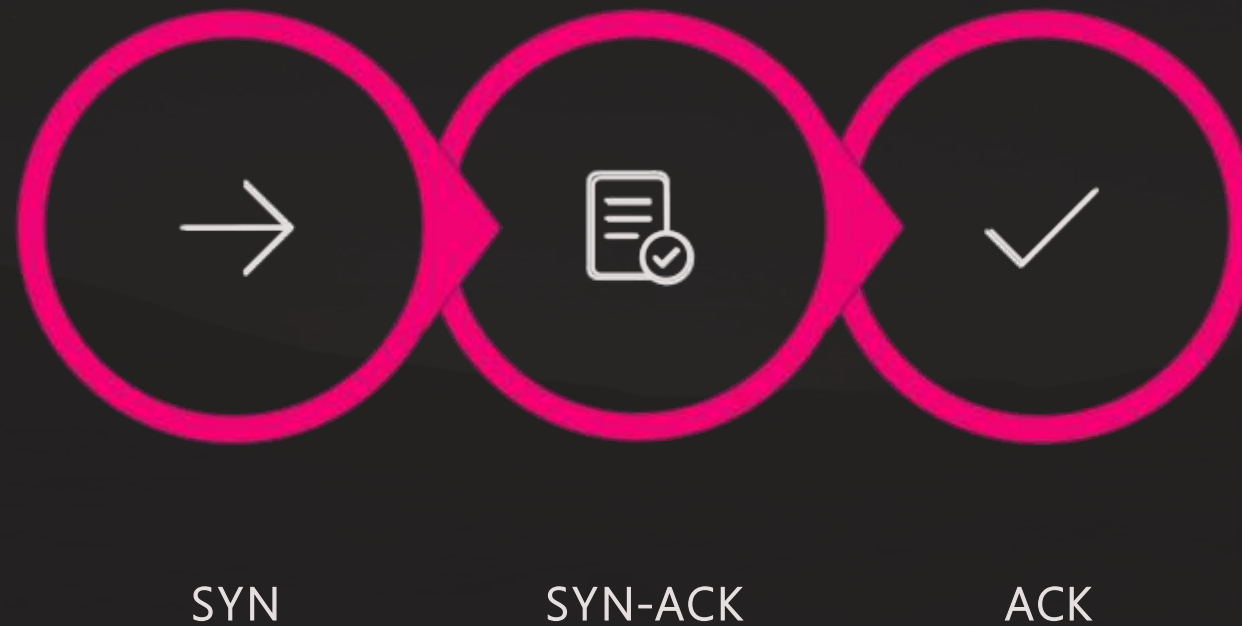
Core Concepts

Focus on secure communication, TCP Handshake, and encryption.

Beginner-Friendly

Designed for students and enthusiasts to grasp complex ideas visually.

The TCP 3-Way Handshake



Witness how a reliable connection is established:

- **SYN:** Client initiates connection.
- **SYN-ACK:** Server acknowledges and responds.
- **ACK:** Client confirms, connection open.

Essential for data exchange integrity.

Encryption in Action



Symmetric Encryption (AES)

Single key for both encryption and decryption.



Asymmetric Encryption (RSA)

Public and private keys for secure data exchange.



Attacker Model

Demonstrates data integrity even if intercepted.

 PROJECT OUTCOME

Bridging Theory and Practice

1 Enhanced Understanding

Transforms abstract concepts into tangible, visual experiences.

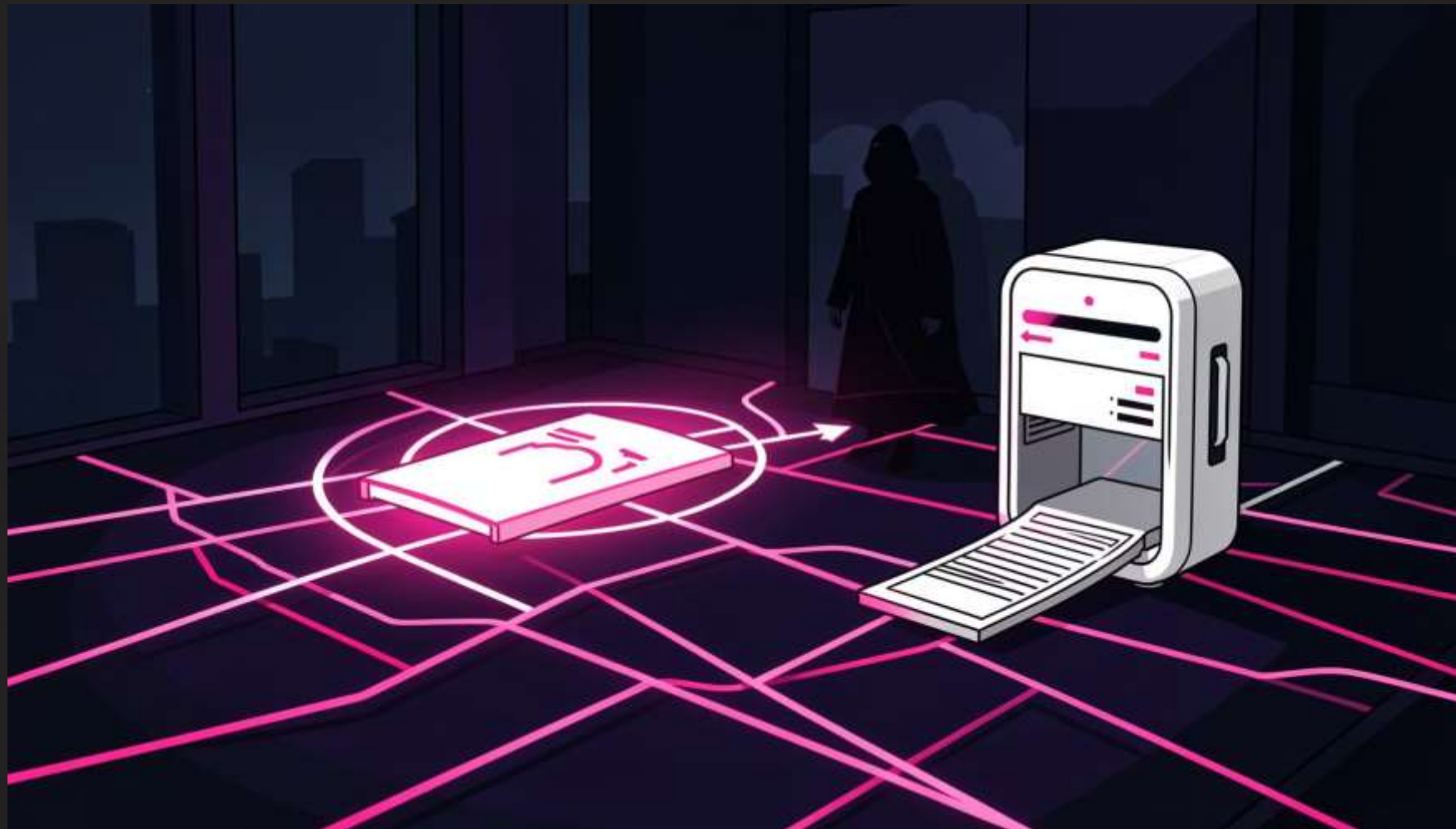
2 Practical Learning

Ideal for visual learners struggling with theoretical explanations.

3 Skill Development

Builds foundational knowledge for future cybersecurity studies.

Visualizing Key Components



Sender & Receiver

Visual representation of communication endpoints.

The Network

How data travels between parties.

The Attacker

Illustrates interception and data security.

 TECHNOLOGY STACK

Tools of the Trade



HTML

Structure for the interactive interface.



CSS

Styling for a clean, intuitive design.



JavaScript

Powers the dynamic simulations and interactions.

Roadmap for Growth

Phase 1: Initial Release

TCP Handshake & Basic Encryption Visualizations.

1

2

Phase 2: Advanced Modules

Implement more complex attacks (e.g., Man-in-the-Middle).

3

Phase 3: Community & Feedback

Gather user input, expand features, and add new topics.